WorldQuant University Applied Data Science Lab Assessment Guide

Python Data Structures, Control Flow, Functions & Statistical Logic Master Blueprint

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 25, 2026

1. WQU Applied Data Science Lab Assessment Topics

Topic / Domain	Key Concepts Tested	Example Python Construct	Passing Tip
Python Data Structures	Lists, Dictionaries, Sets, Tuples	`data = {'a': 1, 'b': 2}`	Key-value lookup is O(1)
Control Flow & Loops	For loops, While loops, If-Else	`[x for x in nums if x > 0]`	List comprehensions save time
Functions & Scope	`def` keywords, return statements	`def calc(a, b): return a + b`	Always return explicit values
Basic Math & Statistics	Mean, Median, Modulo `%`, Sum	`sum(vals) / len(vals)`	Watch out for zero-division errors

2. Strategic Alignment: Syracuse BA/BS + WQU Lab Certificate

Academic Strategy: *Maintaining Syracuse University for BA/BS establishes top institutional prestige, while completing WQU's Applied Data Science Lab provides immediate $0-cost portfolio credentials.*